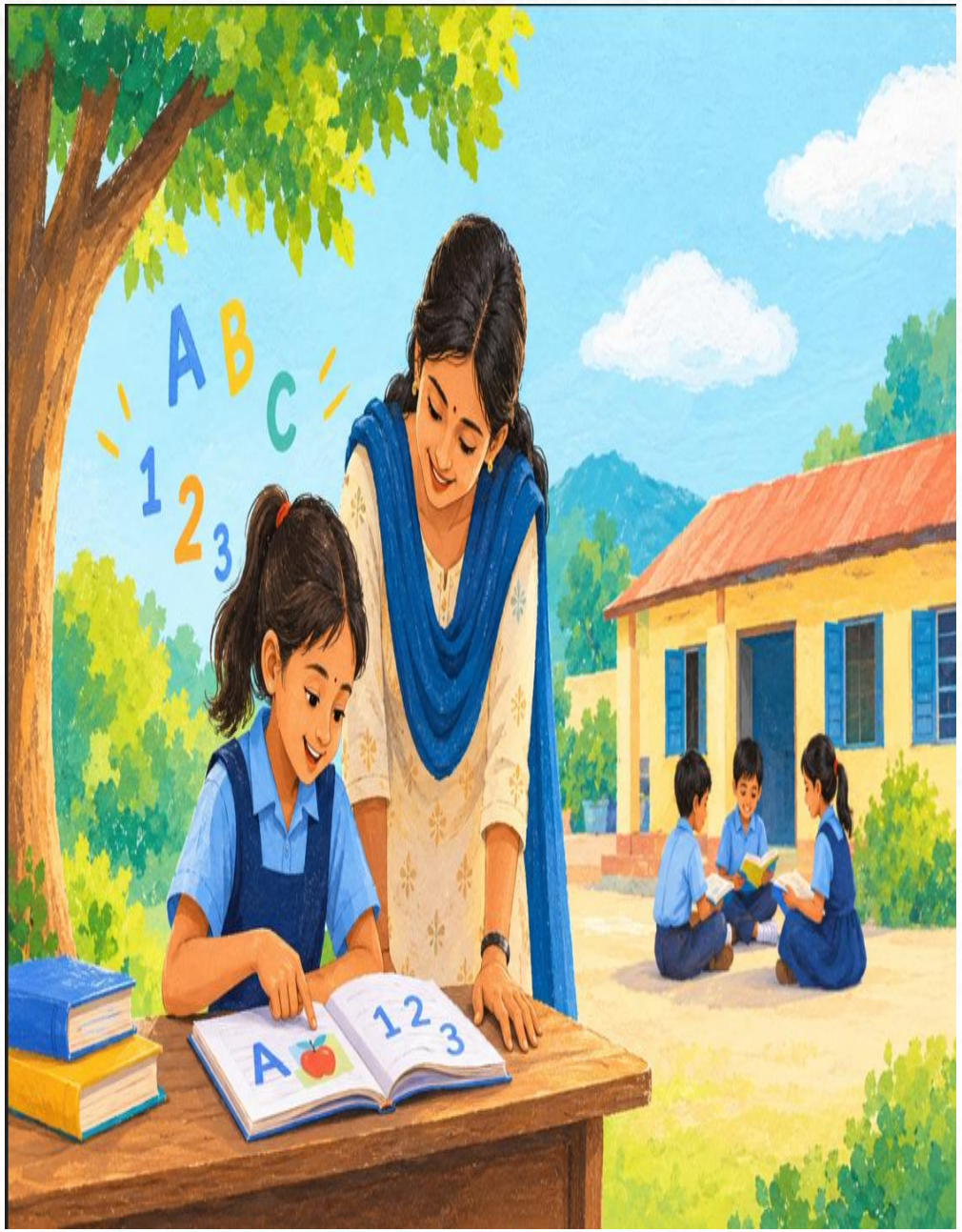
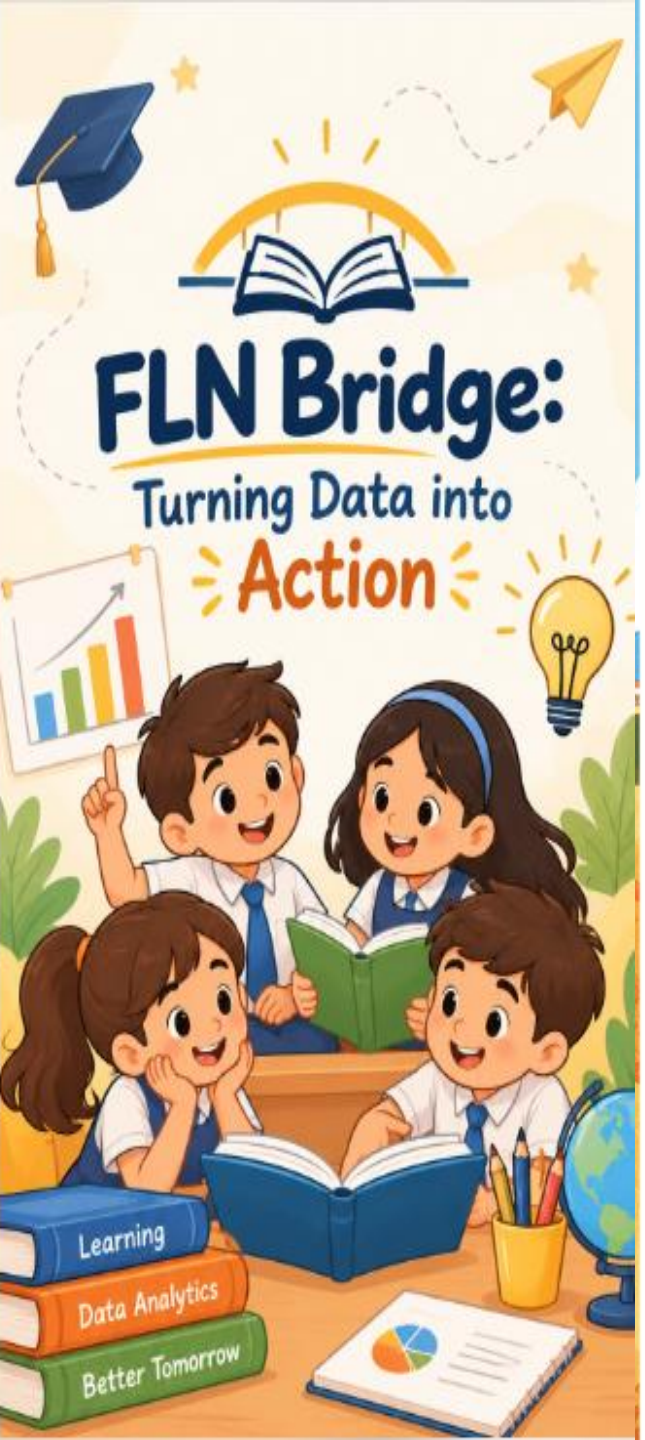


FLN Bridge: Turning Data into Action

From Learning Gaps to Targeted Classroom Solutions

Presenter :
Msiithili Mahajan
Arpita Hanwatkar
Samiksha Shukla
Khushi Krinapure



Contents

01 Context and Core Challenge

02 Ecosystem and Solution

03 Implementation and Impact

01 Context and Core Challenge



FLN Bridge: Techfest 2026 27 Showcase

Project Challenge

Feature the project title “FLN BRIDGE – Smart Learning Gap Detection & Remediation System,” positioned as a flagship entry for The India @ 71/100 Challenge.

Theme and Visual Identity

Pair Theme 2: Foundational Learning Outcomes with team name, IIT Bombay, and Techfest 2026–27, using a premium navy, white, and teal cover alongside a realistic Indian classroom photograph.

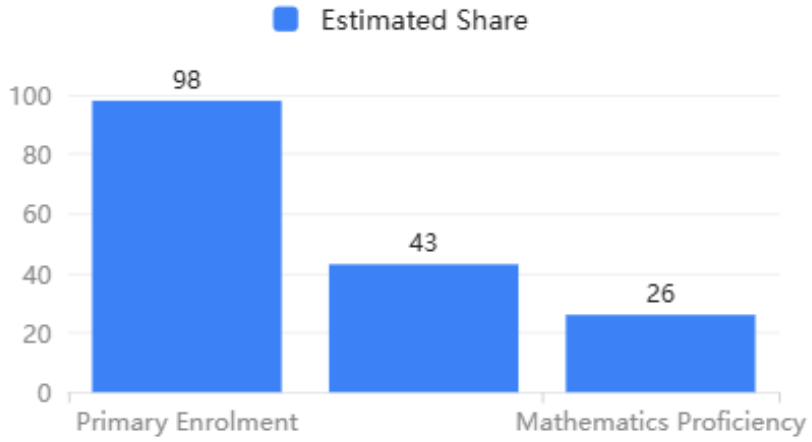


India’ s Learning Gap

Access vs Learning

Average Value	YoY Growth
500	10%

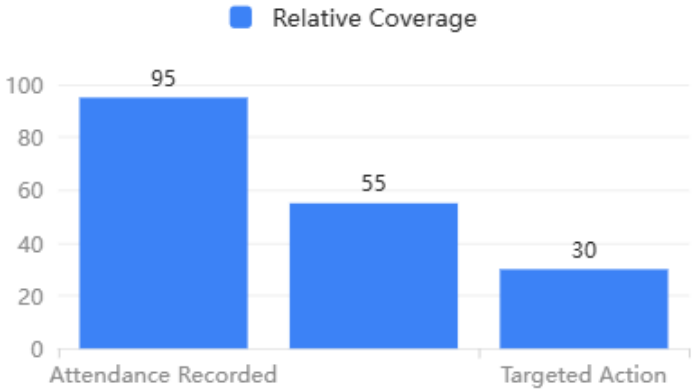
Nearly every child enters primary school, yet far fewer achieve foundational reading and mathematics proficiency, especially in underserved communities.



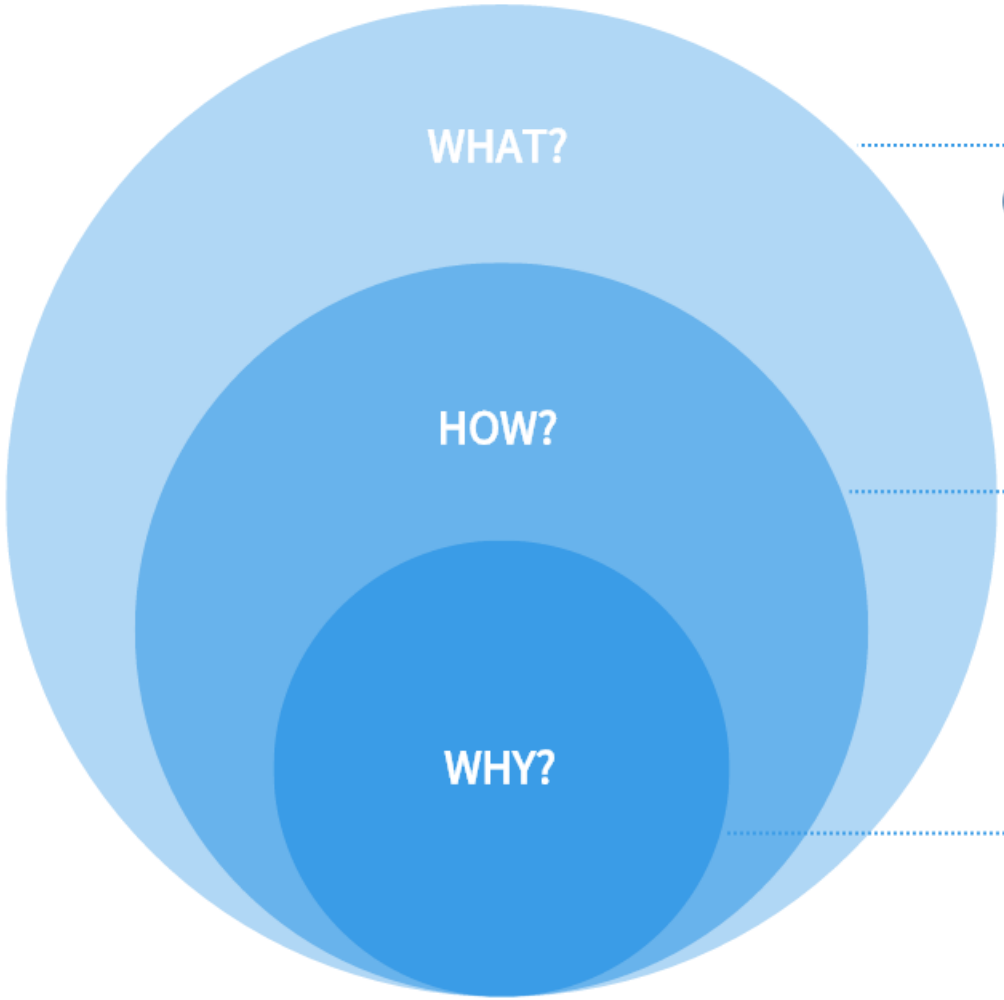
Data to Action

Average Value	YoY Growth
500	10%

Assessment results and attendance records rarely translate into targeted classroom support, leaving persistent learning gaps unaddressed for vulnerable students.



The Challenge to Solve



(Execution)

Phenomenon /Outcome

Learning gaps often widen before they become visible, leaving teachers with limited time to respond. Every child needs timely, individualized support, yet existing school data rarely translates into clear signals for immediate classroom action.

(Method)

Method/Action

Convert attendance, assessment, and classroom data into early learning-gap insights by classifying students, identifying priority needs, recommending targeted remediation, and tracking progress over time so teachers can act before gaps widen.

(Purpose)

Mission/Idea

Enable teachers to make timely, evidence-informed decisions that strengthen learning pathways for every student. The goal is not merely to report performance, but to turn data into practical support, continuous improvement, and equitable opportunity.

02 Ecosystem and Solution



India’ s FLN Ecosystem

NEP 2020: National Reform Framework

NEP 2020 provides India’s overarching education reform framework, aligning foundational learning, curriculum transformation, teacher development, equity, and system-wide implementation priorities.

NIPUN Bharat: Mission for FLN

NIPUN Bharat translates national reform into a focused mission, targeting universal foundational literacy and numeracy through structured milestones, teaching practices, and monitoring.

UDISE+ and PARAKH: Data and Standards

UDISE+ strengthens education data systems, while PARAKH establishes assessment standards, together enabling more consistent measurement of learning outcomes across schools and states.

DIKSHA: Digital Learning Platform

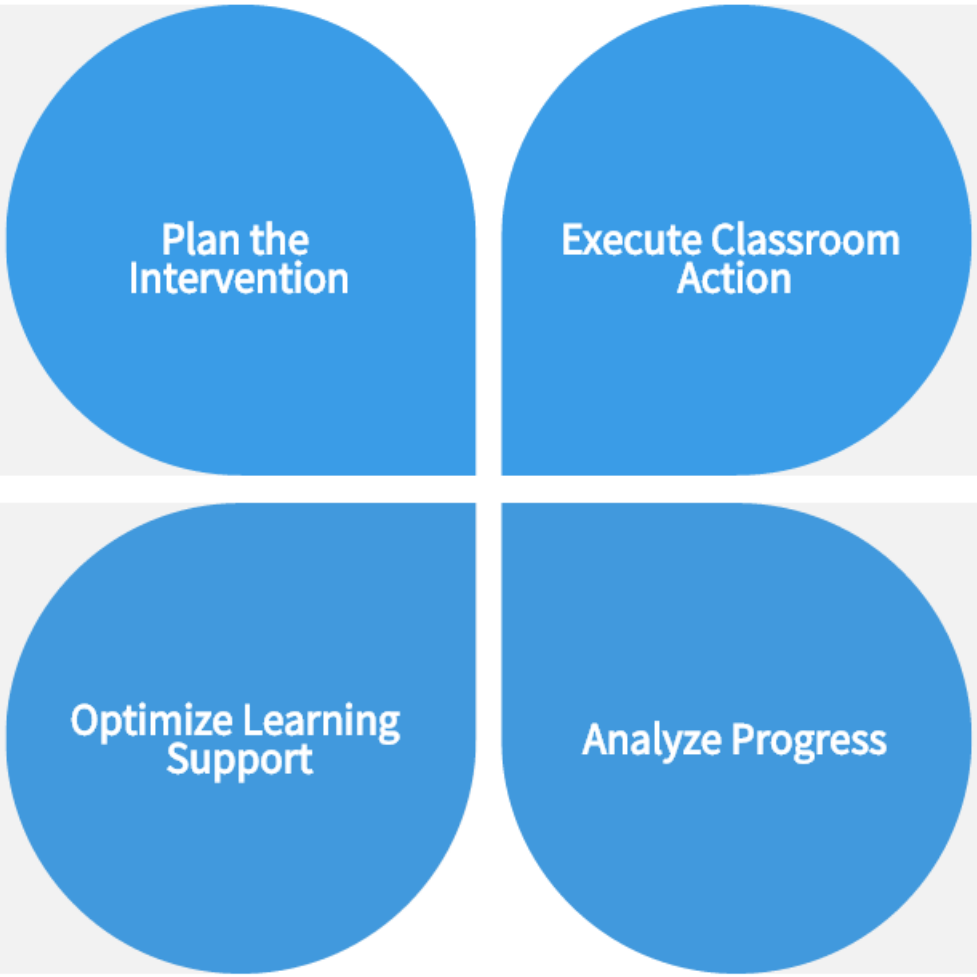
DIKSHA expands access to digital lessons, teacher resources, and curriculum-aligned content, supporting scalable learning delivery across India’s diverse education ecosystem.

Missing Layer: Classroom Decision Support

An integrated classroom-level decision-support layer remains missing, connecting ecosystem data to actionable insights for teachers, school leaders, timely interventions, and improved FLN outcomes.

Introducing FLN Bridge

Map fragmented student data sources, define learning-need categories, and establish intervention priorities that connect FLN Bridge insights to classroom planning.



Capture student data, automatically detect learning gaps, classify learners by need, deliver personalized recommendations, and monitor progress through a teacher-friendly dashboard.

Review detection accuracy, recommendation relevance, and dashboard usability regularly; refine thresholds and workflows so teachers receive clearer, faster, more actionable support.

Compare intervention results across learner groups, identify persistent gaps, and use teacher feedback to improve recommendations without replacing professional judgment.

FLN Bridge Architecture

Existing Education Data

Gather attendance, assessment, learning records, and intervention histories from schools, establishing the trusted input foundation for the FLN Bridge workflow.

Data Collection

Collect data through lightweight, structured channels, preserving consistency across schools while supporting secure transfer, validation, and future integration with existing education systems.

Data Processing

Use Python, Pandas, and NumPy to clean, normalize, and organize incoming records, creating reliable datasets for downstream analysis and model development.

Recommendation Engine

Convert analytical findings into targeted recommendations, prioritizing practical interventions that help educators respond quickly to learner needs and strengthen foundational literacy outcomes.

Teacher Dashboard

Present recommendations through a Streamlit teacher dashboard backed by SQLite, enabling secure, lightweight deployment suited to school environments and daily instructional use.

Analytics Engine

Apply Scikit-learn within the teal intelligence layer to identify learning patterns, surface risks, and translate processed data into actionable insights.

03 Implementation and Impact



How FLN Bridge Works

01

Capture Classroom Data

Teachers enter attendance, assessment results, and student performance data, creating a reliable starting point for understanding classroom learning conditions and needs.

02

Analyze Performance Trends

The system analyzes performance trends, compares learner outcomes, and highlights emerging patterns that reveal where foundational learning gaps require attention.

03

Deliver Targeted Intervention

Teachers apply the recommended activities in class, adapting instruction to learner responses and providing timely, skill-focused support where it matters most.

04

Classify Learner Needs

Students are grouped by specific foundational skill needs, enabling teachers to distinguish reading, numeracy, or other learning priorities for targeted support.

05

Assess Progress and Adapt

Follow-up assessment measures progress, confirms whether learning gaps are closing, and informs the next instructional cycle for continuous instructional improvement.

06

Generate Remedial Recommendations

Based on each learner profile, the system generates practical remedial recommendations; for example, a student struggling with sentence reading receives guided reading practice.

From Pilot to Scale

2024

Phase 1: Research and Discovery

Map learner needs, curriculum expectations, and teacher workflows to establish a practical foundation for evidence-informed design and implementation across schools.

2025

Phase 2: Data and Detection

Develop reliable data-processing and learning-gap detection modules that transform classroom evidence into actionable insights for teachers and education leaders alike.

2026

Phase 3: Dashboard Validation

Build and validate a teacher dashboard that presents clear, timely, and usable information for planning targeted instruction and monitoring learner progress.

2027

Phase 4: School Pilot

Pilot the platform in schools with teachers and education officials, testing usability, implementation conditions, and real-world value across diverse learning contexts.

2028

Phase 5: Evaluation and Expansion

Evaluate outcomes, refine the platform using stakeholder feedback, and expand implementation through evidence-based decisions, stronger support, and scalable delivery models.

Feasibility, Privacy, and Impact

Feasible and Responsible Design

Lightweight tools and local storage enable low-bandwidth deployment, while secure storage, consent mechanisms, and role-based access protect student information throughout implementation.

Stakeholder Benefits

Students receive earlier identification and personalised remediation; teachers reduce manual analysis; schools gain clearer monitoring and evidence-based planning for foundational skills.

Impact Measurement

Track assessment improvement, remedial support delivered, and teacher feedback to verify adoption, prioritize interventions, and refine implementation decisions over time.

Dimension	Implementation Focus	Expected Value
Feasibility	Lightweight tools and local data storage	Reliable use in low-bandwidth environments
Privacy	Secure storage, consent, and role-based access	Protected student information and trusted adoption
Impact	Assessment gains, remedial support, and teacher feedback	Evidence-based intervention and planning

Conclusion: Every Child Can Progress

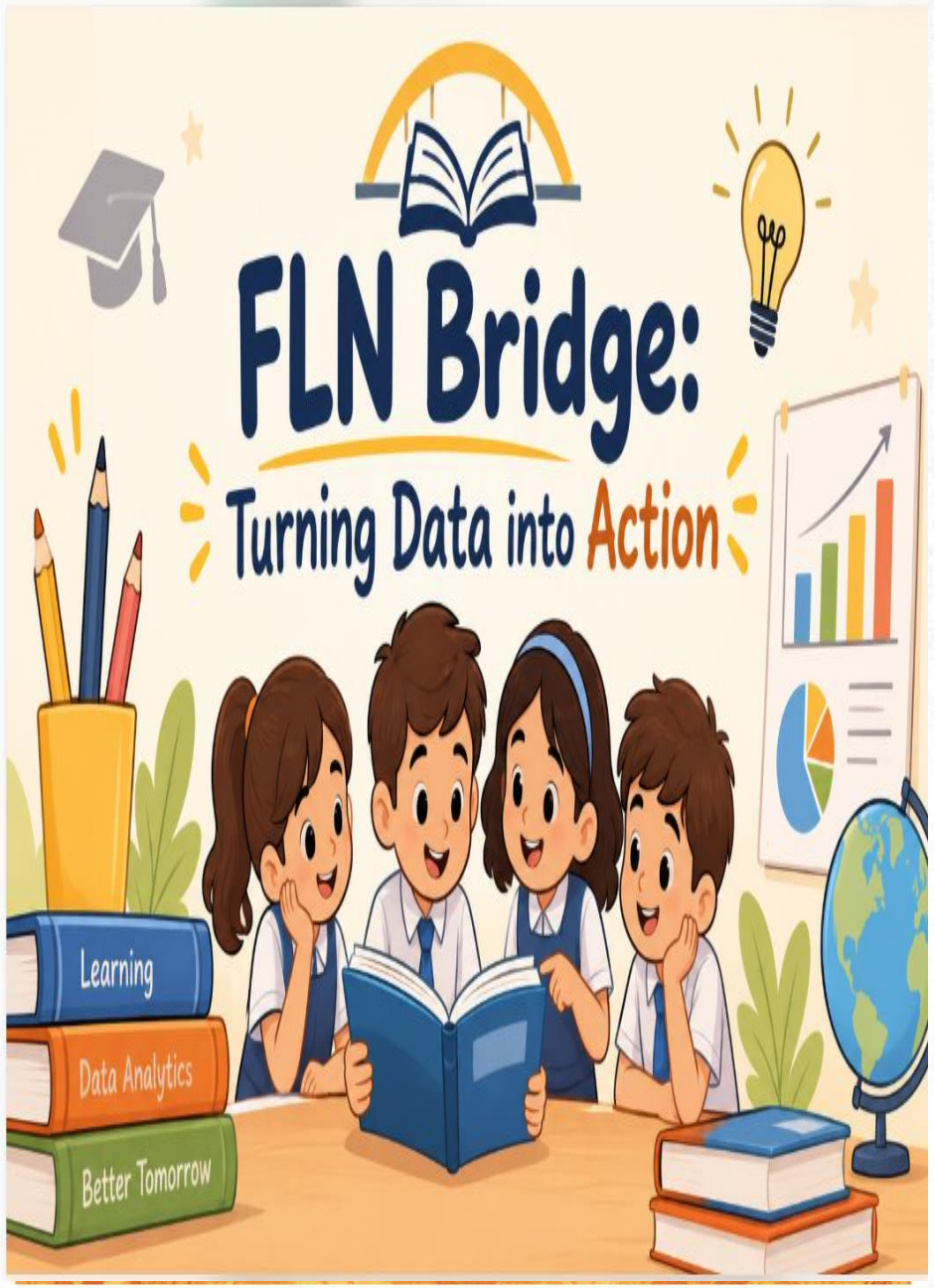


From Data to Classroom Action

FLN Bridge transforms educational data into timely, targeted, and equitable classroom action, helping every primary learner receive support when and where it matters most.

A Promise for Every Learner

Every child deserves learning support according to their individual needs. Thank You. Q&A. Close with a clean navy, white, and teal dashboard or classroom visual.



Thanks

Questions, Reflections, and Next Steps

Presenter :
Msiithili Mahajan
Arpita Hanwatkar
Samiksha Shukla
Khushi Krinapure